Python & R Codes - DSA0602
Class Activity 2

1. Air Quality Monitoring

Python:

```
import pandas as pd import matplotlib.pyplot as plt
df=pd.DataFrame({"PM25":[30,45,60,80,100,120,70,50],"NO2":[15,20,30,40,55,65,
35,25]}) plt.scatter(df["PM25"],df["NO2"],c=df["PM25"],cmap="viridis",s=100)
plt.colorbar(label="PM2.5") plt.title("Air Pollution Hotspots")
plt.xlabel;("PM2.5");plt.ylabel("NO2") plt.show()
```

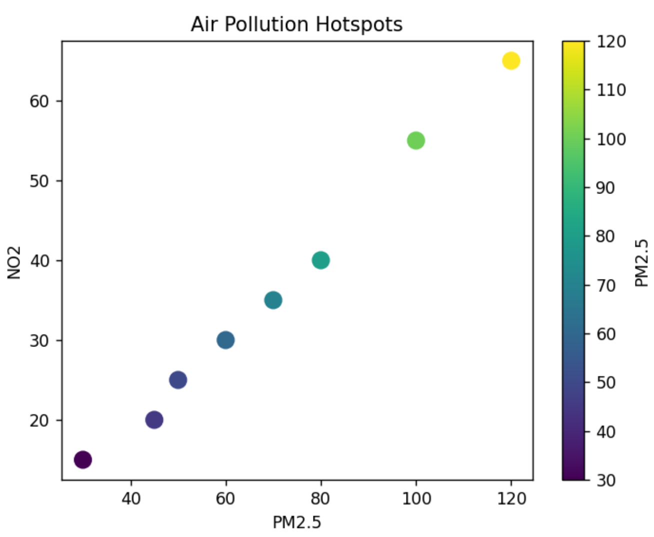
R:

```
library(ggplot2)

df <- data.frame(
PM25=c(30,45,60,80,100,120,70,50),
 NO2=c(15,20,30,40,55,65,35,25)
)

ggplot(df,aes(PM25,NO2,color=PM25))+
geom_point(size=5)+
scale_color_gradient(low="green",high="red")+
ggtitle("Air Pollution Hotspots")
```

Output: A graph window opens displaying the visualization.



2. Earthquake Risk

Python:

```
import matplotlib.pyplot as plt
depth=[5,10,20,35,60,100,180,350,700]
mag=[1,2,3,4,5,6,7,8,9]
plt.plot(depth,mag,marker="o")
plt.xscale("log") plt.title("Earthquake
Distribution (Log Scale)")
plt.xlabel("Depth");plt.ylabel("Magnitude")
plt.grid() plt.show()
```

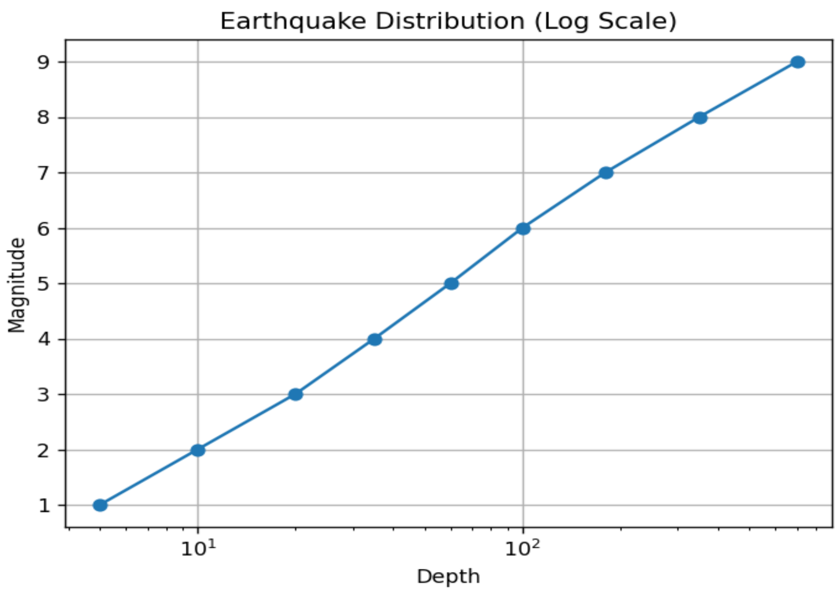
R:

```
library(ggplot2)

df<-data.frame(
Depth=c(5,10,20,35,60,100,180,350,700),
Magnitude=c(1,2,3,4,5,6,7,8,9))

ggplot(df,aes(Depth,Magnitude))+
geom_line()+ geom_point()+
scale_x_log10()
```

Output: A graph window opens displaying the visualization.



3. Wind Direction

Python:

```
import numpy as np import
matplotlib.pyplot as plt
angles=np.deg2rad([0,30,60,90,120,150,180,210,240,270,300,
330]) speed=[10,15,20,12,18,22,16,11,14,19,21,13]
ax=plt.subplot(111,polar=True)
ax.bar(angles,speed,width=np.pi/12) plt.title("Wind Rose")
plt.show()
```

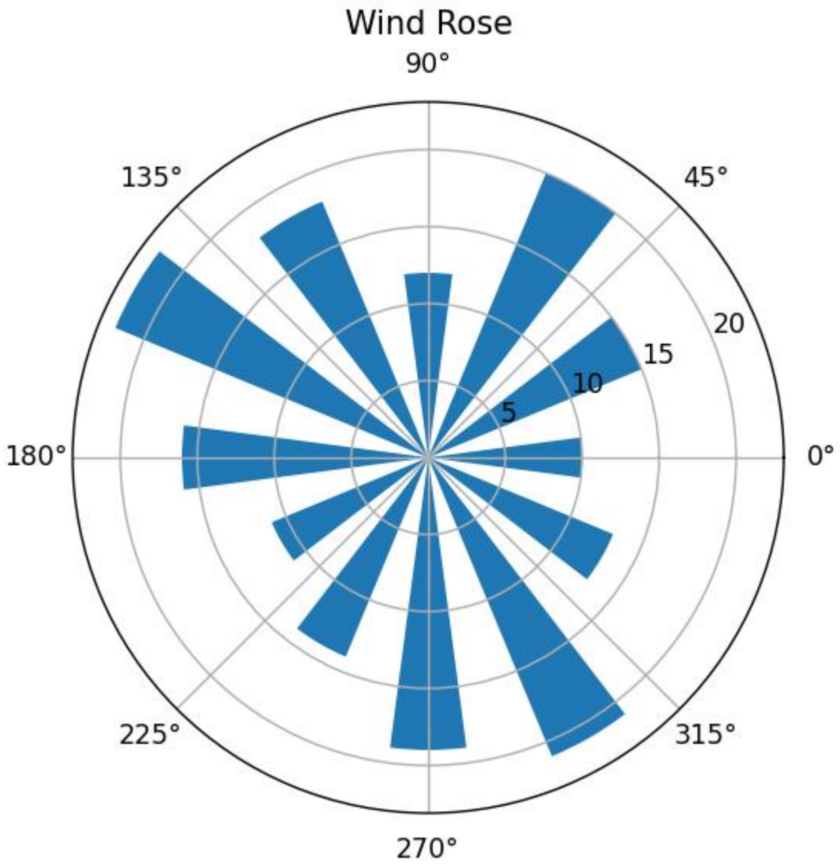
R:

```
library(ggplot2)

df<-data.frame(
Direction=factor(c("N","NE","E","SE","S","SW","W","NW")),
Speed=c(12,18,22,15,10,16,20,14))

ggplot(df,aes(Direction,Speed,fill=Speed))+
geom_bar(stat="identity")+ coord_polar()
```

Output: A graph window opens displaying the visualization.



4. ICU Dashboard

Python:

```
import matplotlib.pyplot as plt
metrics=["HR","SpO2","BP","Tem
p"] values=[120,88,160,39]
colors=["red","orange","red","red"]
plt.bar(metrics,values,color=colors)
plt.title("ICU Dashboard")
plt.show()
```

R:

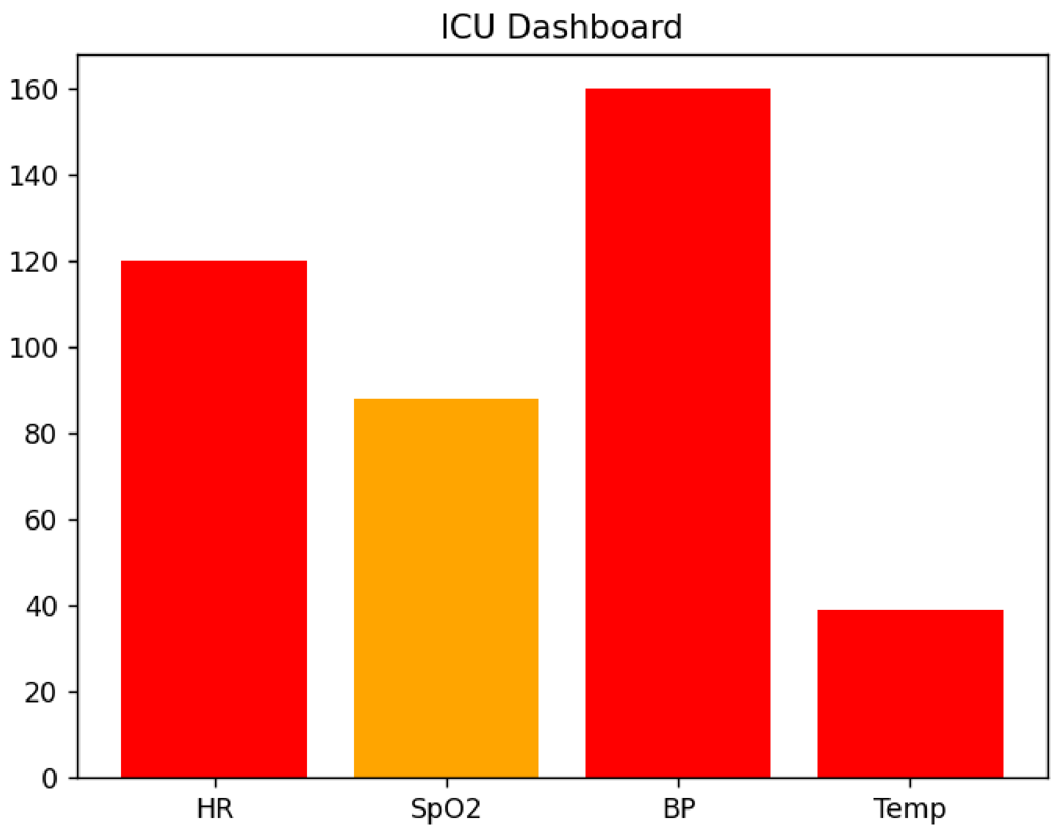
```
library(ggplot2)

df<-data.frame(
Metric=c("HR","SpO2","BP","Temp"),

Value=c(120,88,160,39))

ggplot(df,aes(Metric,Value,fill=Metric))+
geom_col()
```

Output: A graph window opens displaying the visualization.



5. Climate Heatmap

Python:

```
import matplotlib.pyplot as plt
import numpy as np
data=np.random.randn(10,10)
plt.imshow(data,cmap="coolwa
rm") plt.colorbar()
plt.title("Climate Heatmap")
plt.show()
```

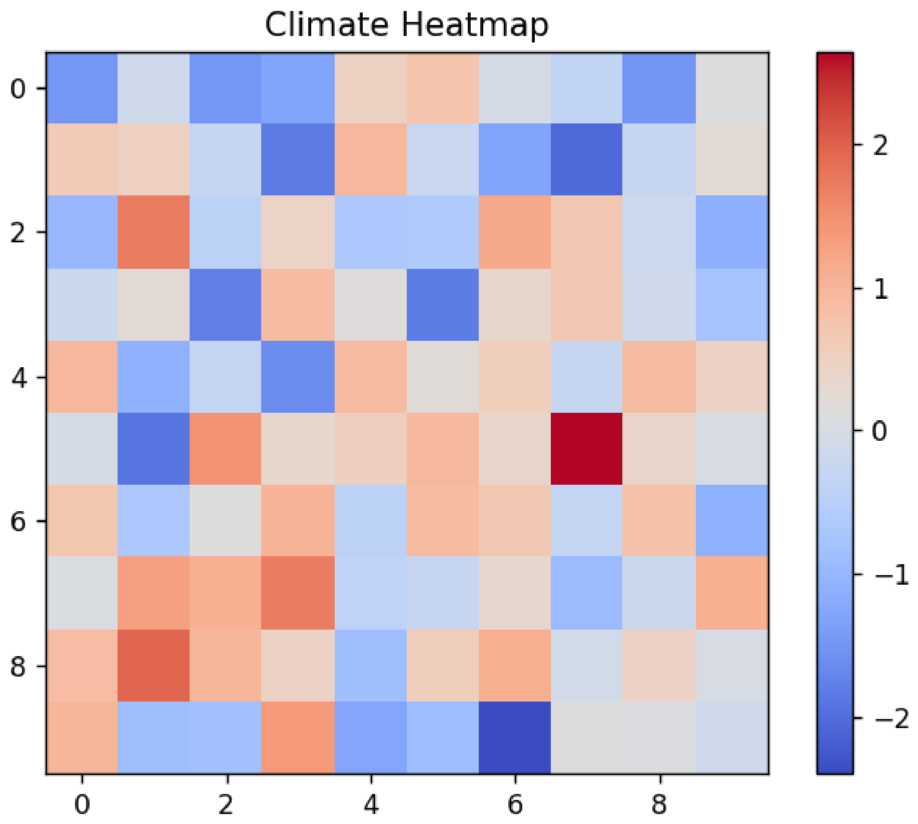
R:

```
library(ggplot2)

set.seed(10) df<-
expand.grid(X=1:10,Y=1:10) df$Temp<-
rnorm(100)

ggplot(df,aes(X,Y,fill=Temp))+
geom_tile()+ scale_fill_gradient2()
```

Output: A graph window opens displaying the visualization.



6. Sales Analytics

Python:

```
import matplotlib.pyplot as plt
categories=["Electronics","Clothing","Food"]
sales=[120,90,150]
plt.bar(categories,sales,color=["red","green","blue"])
plt.title("Sales Analytics") plt.ylabel("Sales")
plt.show()
```

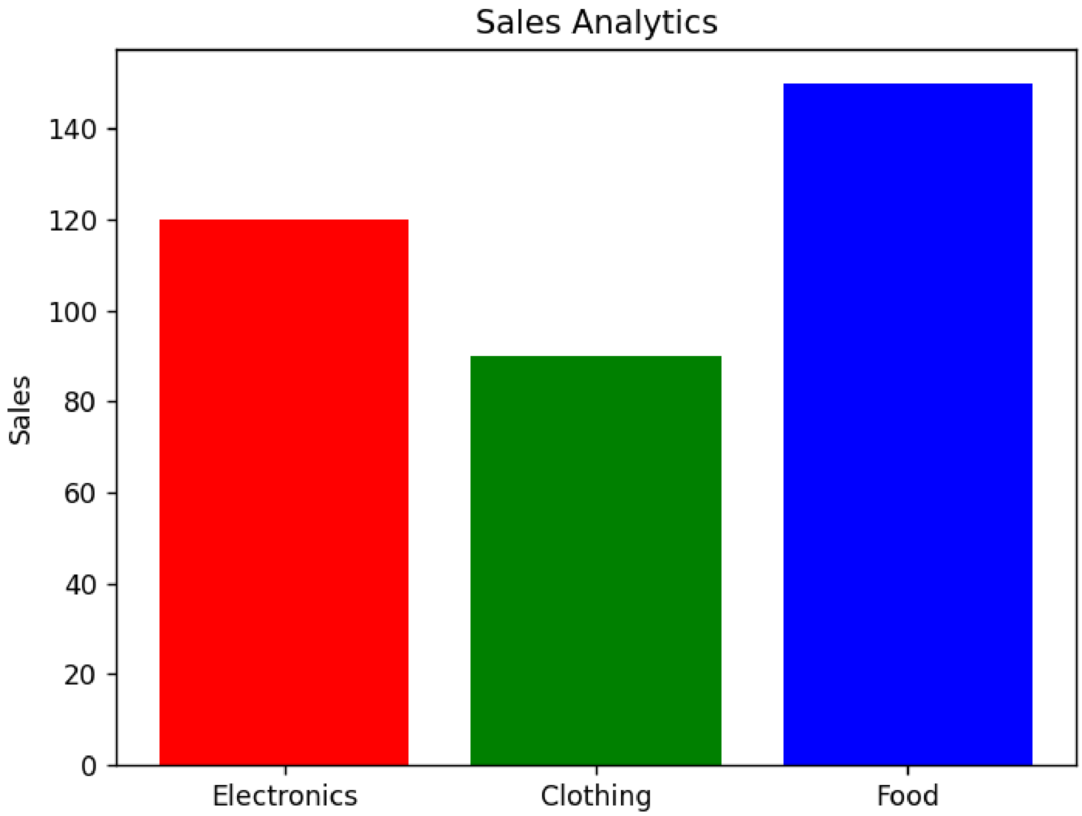
R:

```
library(ggplot2)

df<-data.frame(Category=c("Electronics","Food","Books"),
Sales=c(120,180,150))

ggplot(df,aes(Category,Sales,fill=Category))+
geom_col()
```

Output: A graph window opens displaying the visualization.



7. NDVI Vegetation

Python:

```
import matplotlib.pyplot as plt import
numpy as np ndvi=np.linspace(-
1,1,100).reshape(10,10)
plt.imshow(ndvi,cmap="RdYlGn")
plt.colorbar() plt.title("NDVI
Vegetation") plt.show()
```

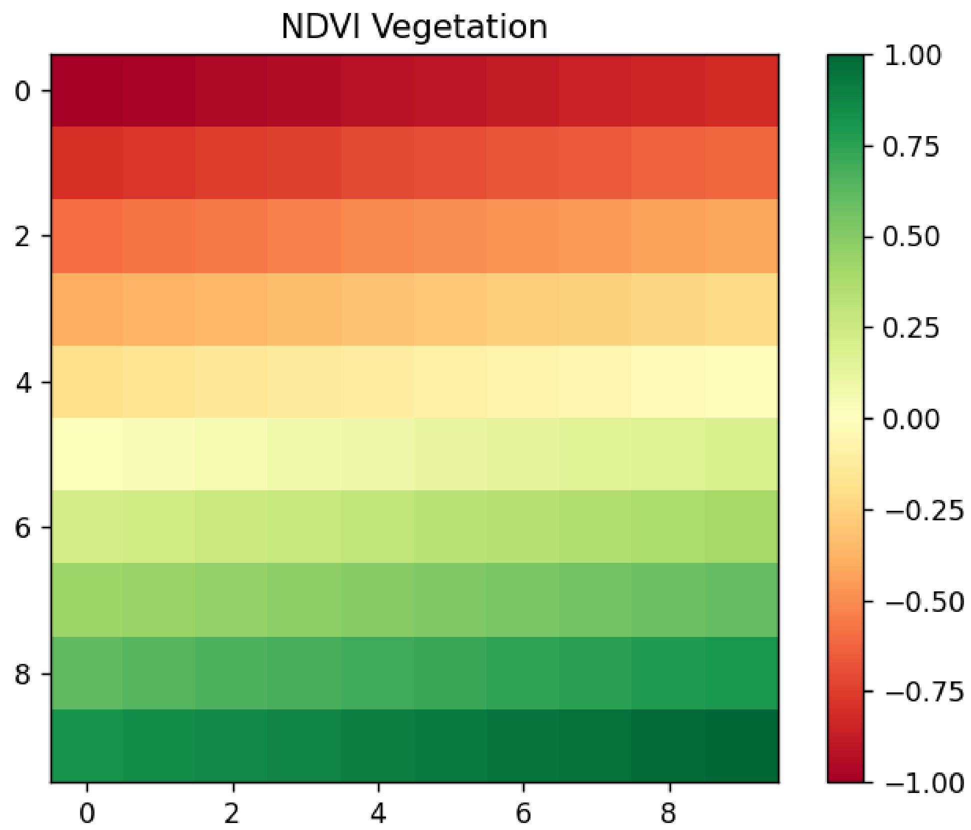
R:

```
library(ggplot2)

df<-expand.grid(X=1:20,Y=1:20) df$NDVI<-
runif(400,-1,1)
```

```
ggplot(df,aes(X,Y,fill=NDVI))+
geom_tile()+
scale_fill_gradient2(low="blue",mid="yellow",high="darkgreen")
```

Output: A graph window opens displaying the visualization.



8. Cryptocurrency

Python:

```
import matplotlib.pyplot as plt
days=[1,2,3,4,5]
btc=[1000,3000,5000,8000,15000]
eth=[100,300,700,1200,2000]
plt.plot(days,btc,label="Bitcoin")
plt.plot(days,eth,label="Ethereum"
) plt.yscale("log") plt.legend()
plt.title("Crypto Prices")
plt.show()
```

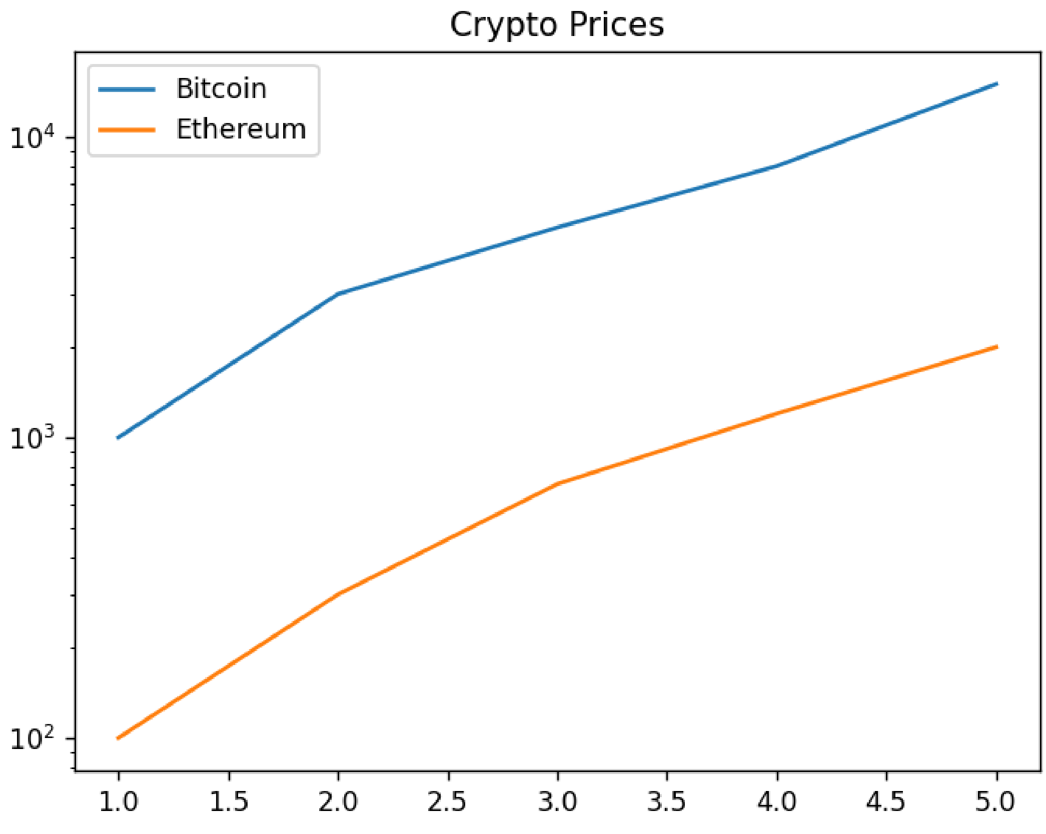
R:

```
library(ggplot2)

df<-data.frame(Day=1:5,
Bitcoin=c(1000,3000,5000,9000,15000))

ggplot(df,aes(Day,Bitcoin))+
geom_line()+ scale_y_log10()
```

Output: A graph window opens displaying the visualization.



9. Marathon

Python:

```
import matplotlib.pyplot as plt
distance=[5,10,15,20,25,30,35,42]
speed=[12,11,10,10.5,9.8,9.5,9.2,8.8]
plt.scatter(distance,speed,c=speed,cmap="plasma",s=1
00) plt.colorbar(label="Speed") plt.xlabel("Distance")
plt.ylabel("Speed") plt.title("Runner Performance")
plt.show()
```

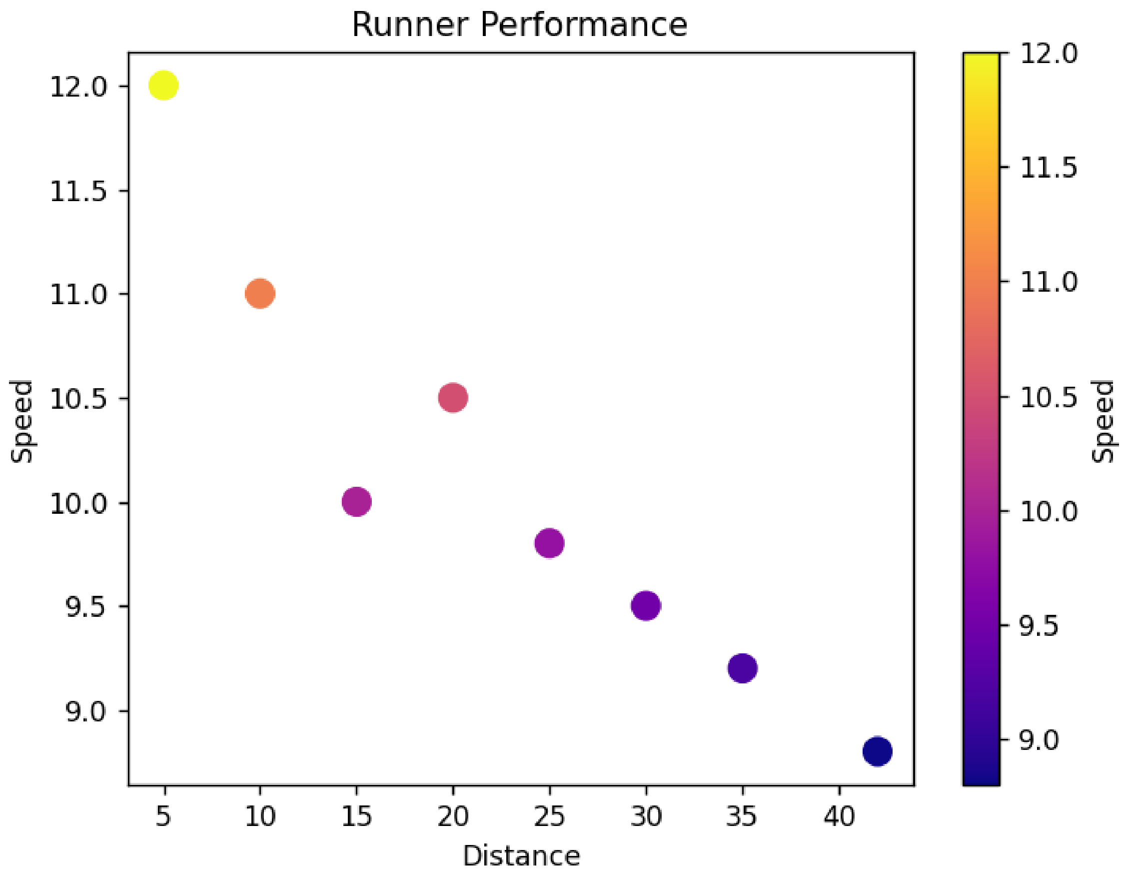
R:

```
library(ggplot2)

df<-data.frame(Distance=c(5,10,15,20,25,30,35,42),
Speed=c(12,11,10,10.5,9.8,9.5,9.2,8.8))

ggplot(df,aes(Distance,Speed,color=Speed))+
geom_point(size=4)
```

Output: A graph window opens displaying the visualization.



10. COVID Dashboard

Python:

```
import matplotlib.pyplot as plt
district=["A","B","C","D"]
vacc=[70,85,90,65]
plt.bar(district,vacc,color="skyblue")
plt.ylim(0,100) plt.title("COVID
```

Vaccination Rates") plt.ylabel("%
Vaccinated") plt.show()

R:

library(ggplot2)

df<-data.frame(District=c("A","B","C","D"),
Vaccination=c(70,85,90,65))

ggplot(df,aes(District,Vaccination,fill=District))+
geom_col()+ ylim(0,100)

Output: A graph window opens displaying the visualization.